

Cryogenic Valve Fatigue Paper

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Abstract

Four 316L valves were cycled between ambient and cryogenic conditions to distinguish reversible seat leakage from a stable failure threshold. The analysis retains censored cycles and does not infer failure before the measured criterion is met.

1. Introduction

Thermal cycle tests frequently confound transient seal contraction with permanent seat damage. We define failure from repeated leakage, not from a single excursion [1].

Criterion

Failure threshold was reached when Q_{leak} exceeded 0.80 mg s^{-1} in three consecutive measurements.

1.1 Test matrix

CV-11 through CV-14 used identical stems and 316L bodies. Seat batches differed, allowing a bounded comparison without claiming a population estimate.

Instrumentation

Pressure difference Δp was logged independently of impedance Z , and uncertainty was propagated before threshold classification.

$$Q_{leak} = \Delta p / Z$$

2. Endurance protocol

Each thermal cycle held the valve at -196 °C for 8 min, returned it to 22 °C, and paused until stem temperature stabilized within ± 0.4 °C.

2.1 Interruptions

Planned inspections occurred at 2,500-cycle intervals. Unplanned interruption codes were retained to prevent maintenance from appearing as continuous exposure.

$$N_f = A(\Delta\sigma)^{-m}$$

2.2 Measurement checks

Blank fixture leakage remained below 0.05 mg s^{-1} . CV-14 showed a reversible spike after cycle 7,500 but returned below the threshold after equilibration.

Cross-page observation

The endurance sequence was intentionally interrupted after CV-12 produced an asymmetric stem trace. The valve remained sealed, the fixture was photographed, and the next cycle began only after

the instrumentation lead confirmed that the displacement channel had not saturated. This interruption belongs to the same observation and must not be read as a separate test event.

3. Results

CV-11 and CV-13 remained below the failure threshold through 20,000 cycles. CV-12 crossed the threshold at 16,250 cycles; CV-14 remained censored at 18,000 cycles.

Valve	Cycles	Q _{leak}	Status
CV-11	20,000	0.31	censored
CV-12	16,250	0.94	failed
CV-13	20,000	0.28	censored
CV-14	18,000	0.62	censored

3.1 Leakage trend

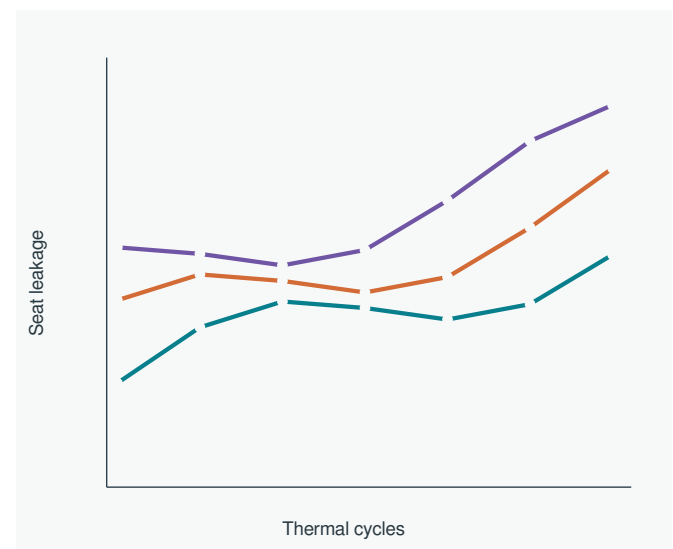


Figure 1. Seat leakage by thermal cycle. The dashed conceptual threshold is 0.80 mg s⁻¹.

Footnote 1. Censored denotes test termination before a repeated threshold crossing.

4. Fractography

Post-test inspection found a shallow circumferential mark on CV-12. The mark coincided with the leakage increase but does not by itself establish a crack-growth mechanism.

4.1 Model comparison

The stress-range model ranked CV-12 as the shortest-lived unit under every admissible exponent m . Confidence bounds remained wide because only one valve met the repeated threshold.

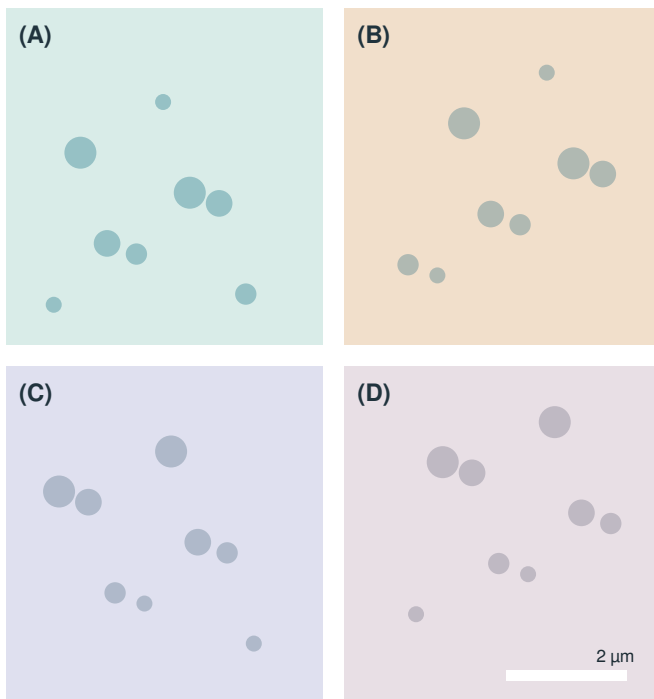


Figure 2. Seat surfaces for CV-11 to CV-14 after testing. Panel letters follow valve order.

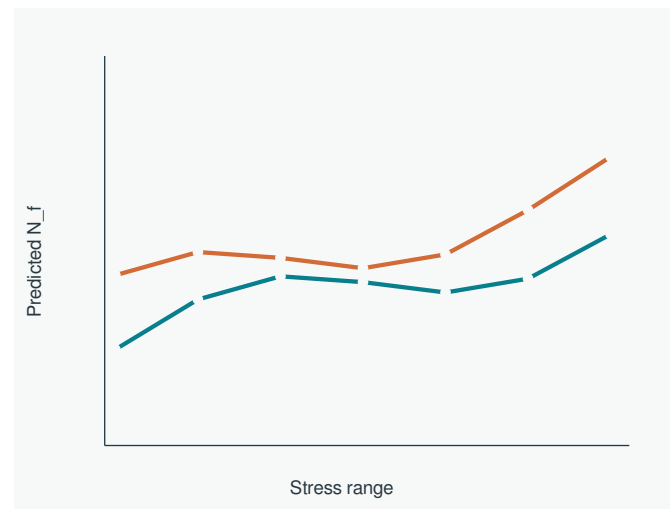


Figure 3. Model sensitivity to $\Delta\sigma$. Lines are conditional fits, not independent observations.

5. Conclusions

Repeated leakage distinguished a stable CV-12 failure from transient cryogenic excursions. The small sample supports procedural conclusions about thresholding but not a general failure-rate estimate for 316L valves.

References

- [1] Chen L. Reversible seal contraction during cryogenic service. Reliability Letters 11:21-30. 10.5555/cryo.2026.011.
- [2] Nakatani E, Moore S. Censoring valve endurance tests. Mechanical Evidence 7:91-104.
- [3] Ivers P. Impedance checks for low leakage measurements. Instrument Practice 4:8-17.
- [4] Moore S. Threshold rules under interrupted exposure. Test Design Quarterly 18:55-63.
- [5] Nakatani E. Reporting single-unit material marks. Fracture Notes 2:14-19.